

STAT 2020 Discussion Slides

Week 13: Hypothesis Testing for a Mean (σ Known)

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Agenda

- Practice hypothesis testing for μ when σ is known
- Use the 3-step testing framework clearly and consistently
- Work both left-tailed and right-tailed tests
- Explain rejection regions and their link to significance level α
- Compute and interpret p-values with worked examples
- Attendance check

Test Setup: Mean with Known σ

We test a population mean with known standard deviation σ .

Typical hypotheses:

$$H_0 : \mu = \mu_0 \quad \text{vs.} \quad H_a : \mu < \mu_0, \mu > \mu_0, \text{ or } \mu \neq \mu_0.$$

Test statistic:

$$Z^* = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}.$$

Conditions used in this course:

- Normal population: valid for any n .
- Non-normal population: use when $n \geq 30$.

3-Step Testing Template

Step 1 (Setup):

- State H_0 , H_a , and significance level α .
- Identify tail type (left or right in today's focus).

Step 2 (Decision Rule):

- Find critical value and rejection region.
- Right-tail: reject if $Z^* \geq z_\alpha$.
- Left-tail: reject if $Z^* \leq -z_\alpha$.

Step 3 (Compute + Conclude):

- Compute observed z^* .
- Compare with rejection rule (or p-value) and conclude in context.

Example A: Left-Tailed Test

A filling process targets $\mu_0 = 500$ ml. Known $\sigma = 12$, sample size $n = 36$, sample mean $\bar{X} = 495$. Test at $\alpha = 0.05$:

$$H_0 : \mu = 500, \quad H_a : \mu < 500.$$

Step 2: Left-tail critical value is $-z_{0.05} = -1.645$. Reject if $Z^* \leq -1.645$.

Step 3:

$$z^* = \frac{495 - 500}{12/\sqrt{36}} = \frac{-5}{2} = -2.50.$$

Since $-2.50 \leq -1.645$, reject H_0 .

Conclusion: evidence suggests the true mean is below 500 ml.

Example B: Right-Tailed Test

A process has benchmark $\mu_0 = 100$. Known $\sigma = 8$, $n = 64$, $\bar{X} = 103$. Test at $\alpha = 0.01$:

$$H_0 : \mu = 100, \quad H_a : \mu > 100.$$

Step 2: Right-tail critical value is $z_{0.01} = 2.326$. Reject if $Z^* \geq 2.326$.

Step 3:

$$z^* = \frac{103 - 100}{8/\sqrt{64}} = \frac{3}{1} = 3.00.$$

Since $3.00 \geq 2.326$, reject H_0 .

Conclusion: evidence suggests the true mean is above 100.

What Is a Rejection Region?

Rejection region = values of the test statistic that lead us to reject H_0 .

It is chosen so that, under H_0 :

$$P(\text{falling in rejection region}) = \alpha.$$

Interpretation of α :

- α controls Type I error (false alarm) probability.
- Smaller α : harder to reject H_0 .
- Larger α : easier to reject H_0 .

Rejection Rules by Tail Type

For $Z^* \sim N(0, 1)$ under H_0 :

Right-tailed $H_a : \mu > \mu_0$:

Reject H_0 if $z^* \geq z_\alpha$.

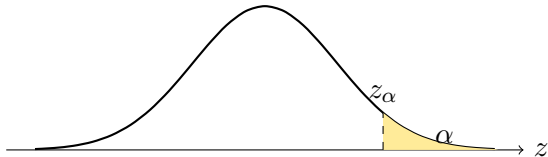
Left-tailed $H_a : \mu < \mu_0$:

Reject H_0 if $z^* \leq -z_\alpha$.

Two-sided (quick reminder) $H_a : \mu \neq \mu_0$:

Reject H_0 if $|z^*| \geq z_{\alpha/2}$.

Sketch: Right-Tail Rejection Region



Right-tailed test at $\alpha = 0.05$: reject when z^* lands in shaded tail.

Definition

What Is a p-Value?

The p-value is the probability, assuming H_0 is true, of getting a test statistic at least as extreme as what we observed.

For observed z^* :

$$\text{Right-tail: } p = P(Z^* \geq z^*), \quad \text{Left-tail: } p = P(Z^* \leq z^*).$$

$$\text{Two-sided: } p = 2P(Z^* \geq |z^*|).$$

Important: p-value is **not** $P(H_0 \text{ is true})$.

How p-Value Connects to α

Equivalent decision rule:

- If $p \leq \alpha$: reject H_0 .
- If $p > \alpha$: fail to reject H_0 .

Interpretation:

- Smaller p-value $\Rightarrow$ stronger evidence against H_0 .
- p-value is the smallest significance level at which H_0 would be rejected.

Worked Example 1 (Right Tail)

p-Value Example 1: Right-Tailed

Test:

$$H_0 : \mu = 50, \quad H_a : \mu > 50.$$

Given $\sigma = 10$, $n = 25$, $\bar{X} = 54$:

$$z^* = \frac{54 - 50}{10/\sqrt{25}} = 2.00.$$

Right-tail p-value:

$$p = P(Z^* \geq 2.00) = 0.0228.$$

Decisions:

- At $\alpha = 0.05$: reject H_0 .
- At $\alpha = 0.01$: fail to reject H_0 .

p-Value Example 2: Left-Tailed

Test:

$$H_0 : \mu = 70, \quad H_a : \mu < 70.$$

Given $\sigma = 15$, $n = 25$, $\bar{X} = 65.8$:

$$z^* = \frac{65.8 - 70}{15/\sqrt{25}} = \frac{-4.2}{3} = -1.40.$$

Left-tail p-value:

$$p = P(Z^* \leq -1.40) = 0.0808.$$

At $\alpha = 0.05$, $p > \alpha$, so fail to reject H_0 .

Worked Example 3 (Match with Rejection Region)

p-Value Example 3: Same Conclusion as Critical-Value Method

Right-tailed test with $\alpha = 0.10$, observed $z^* = 1.58$.

p-value method:

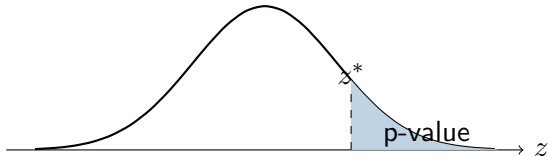
$$p = P(Z^* \geq 1.58) = 0.0571 < 0.10 \Rightarrow \text{reject } H_0.$$

Critical-value method:

$$z_{0.10} = 1.282, \quad 1.58 \geq 1.282 \Rightarrow \text{reject } H_0.$$

Both methods must agree when done correctly.

Sketch: p-Value as Tail Area



For a right-tailed test, p-value is the shaded area to the right of z^* .

Practice 1

Practice 1: Left-Tailed Decision

Given $\mu_0 = 120$, $\sigma = 18$, $n = 36$, $\bar{X} = 114$, $\alpha = 0.05$.

Test $H_0 : \mu = 120$ vs $H_a : \mu < 120$.

Practice 1

Practice 1: Left-Tailed Decision

Given $\mu_0 = 120$, $\sigma = 18$, $n = 36$, $\bar{X} = 114$, $\alpha = 0.05$.

Test $H_0 : \mu = 120$ vs $H_a : \mu < 120$.

$$z^* = \frac{114 - 120}{18/\sqrt{6}} = -2.00, \quad -z_{0.05} = -1.645.$$

Since $-2.00 \leq -1.645$, reject H_0 .

Practice 2

Practice 2: Right-Tailed p-Value

Given $\mu_0 = 40$, $\sigma = 5$, $n = 25$, $\bar{X} = 41.5$, $\alpha = 0.05$.

Test $H_0 : \mu = 40$ vs $H_a : \mu > 40$.

Practice 2

Practice 2: Right-Tailed p-Value

Given $\mu_0 = 40$, $\sigma = 5$, $n = 25$, $\bar{X} = 41.5$, $\alpha = 0.05$.

Test $H_0 : \mu = 40$ vs $H_a : \mu > 40$.

$$z^* = \frac{41.5 - 40}{5/5} = 1.50, \quad p = P(Z^* \geq 1.50) = 0.0668.$$

Since $p > 0.05$, fail to reject H_0 .

Practice 3

Practice 3: Interpret a p-Value

A right-tailed test reports $p = 0.032$.

1. Decision at $\alpha = 0.05$?
2. Decision at $\alpha = 0.01$?

Practice 3

Practice 3: Interpret a p-Value

A right-tailed test reports $p = 0.032$.

1. Decision at $\alpha = 0.05$?
2. Decision at $\alpha = 0.01$?

Answer:

- At $\alpha = 0.05$: reject H_0 .
- At $\alpha = 0.01$: fail to reject H_0 .

Take-Home Messages

- Follow the 3-step hypothesis-testing structure every time.
- Rejection region is chosen from α , before seeing data.
- p-value is a tail probability under H_0 , not a probability that H_0 is true.
- Critical-value and p-value methods should give the same conclusion.
- State conclusions in context (about μ).

In-Class Attendance Check (Quick)

Test at $\alpha = 0.05$:

$$H_0 : \mu = 200, \quad H_a : \mu > 200.$$

Given $\sigma = 20$, $n = 64$, $\bar{X} = 204.5$.

- 1) Compute z^* .
- 2) Find the p-value.
- 3) State the decision.

Please submit: Full Name; Section Number; Your Answer

In-Class Attendance Check (Quick)

Test at $\alpha = 0.05$:

$$H_0 : \mu = 200, \quad H_a : \mu > 200.$$

Given $\sigma = 20$, $n = 64$, $\bar{X} = 204.5$.

- 1) Compute z^* .
- 2) Find the p-value.
- 3) State the decision.

Please submit: Full Name; Section Number; Your Answer

Answer:

$$z^* = \frac{204.5 - 200}{20/8} = 1.80, \quad p = P(Z^* \geq 1.80) = 0.0359.$$

Since $0.0359 < 0.05$, reject H_0 .